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RESEARCH ARTICLES

Episcleral brachytherapy for uveal melanoma: analysis of 136 cases

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Abstract
Purpose To assess the results of I-125 episcleral brachytherapy (EB) in uveal melanoma: tumour control, visual acuity (VA), eye preservation and survival.
Patients Prospective and consecutive study of patients with a diagnosis of uveal melanoma at the Ocular Oncology Unit in the Valladolid University Teaching Hospital treated with EB between September 1997 and June 2008. Ocular examination and extraocular and systemic extension data were registered in a database at the time of the diagnosis and during the follow-up.
Results Among a total of 310 patients diagnosed between September 1997 and June 2008, 136 were treated with EB (mean age, 58.3). Mean follow-up was 55.3 months. As for tumour type, 66.9% were nodular and 39% mushroom shaped. With respect to size, 80.9% were medium, 7.4% small and 11.8% large. After 4.6 years of follow-up, tumours were controlled in 97.1%, with a 55.1% reduction in mean height; only 2.9% of patients showed recurrence. VA was maintained in 16.2% of patients and 17.6% showed an increase; 33% had retinopathy and 14.6% optic neuropathy. Only 5.1% of patients underwent enucleation due to complications and there has been 1 melanoma-related death to date.
Conclusions I-125 EB is effective in tumour control, allowing preservation of the eye and useful visual function for the majority of patients.

Keywords Intraocular tumours · Uveal melanoma · Episcleral brachytherapy

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Introduction

Choroidal melanoma is the most frequent primary malignant intraocular tumour in adults, with an estimated incidence of 4–5 cases per million inhabitants in the USA [1]. Enucleation was the treatment of choice for decades, but in the second half of the 20th century, this radical treatment was shown not to improve patient survival compared with other conservative therapies [2, 3].

The goals of such conservative therapies are to destroy the tumour while causing the least possible side effects, to conserve the eye and visual function, and–if possible–to improve patient survival. However, the latter goal has been shown to be independent of the treatment method applied [4–6].

Episcleral brachytherapy (EB) is now one of the most widely used methods for treatment of uveal melanomas. Different isotopes have been used over time, iodine 125 (^{125}I) and ruthenium 106 (^{106}Ru) being the ones most often used at present. ^{125}I has a greater penetration capacity (tumours of up to 10 mm in height) than ^{106}Ru (up to 5 mm in height) [7].

Table 1 Classification of uveal melanoma

COMS	Height	Base	TNM	Height	Base
Small	≤3 mm	5–10 mm	T1	≤2.5 mm	≤10 mm
Medium	3.1–10 mm	≤16 mm	T2	2.5–10 mm	>10–16 mm
Large	≥10.1 mm	≥16 mm	T3	>10 mm	>16 mm
			T4	>10mm	>16 mm with extraocular extension

The dose that must be reached in the tumour apex for the treatment to be considered effective is 85 Gy [8].

Not only does the radiation affect the tumour, it affects adjacent ocular structures as well; side effects that are acute (retinal detachment, inflammation and intraocular haemorrhage) and long term (retinopathy and neuropathy from radiation) thus appear. Complications arising from episcleral plaque brachytherapy depend on the interaction of the radiation with the tissues next to the plaque [9, 10].

EB is chosen over enucleation in uveal melanoma treatment to avoid losing an eye with visual power. Several studies have demonstrated that there are no statistically significant differences in survival of patients treated with EB over those enucleated, as long as tumour size and extension are within the indications for this treatment [9, 11–14]. The degree of visual acuity (VA) that can be conserved in an eye undergoing EB depends on tumour location and size, treatment given and patient characteristics [15]. The purpose of this study was to evaluate our EB results with ^{125}I in uveal melanoma with respect to tumour control, side effects, eye conservation and survival; and to compare our results with those of previously published studies.

Patients, material and methods

Patients, inclusion criteria and data register

This prospective and consecutive study includes patients diagnosed with choroidal melanoma treated with EB with ^{125}I between September 1997 and June 2008 in the Ocular Oncology Unit of the University Teaching Hospital in Valladolid (Spain).

All patients received a complete ocular exam including determination of the best VA with optical correction, anterior pole biomicroscopy, fundus eye examination, and B-mode ultrasound with vector A, measurement of tumour diameter. Other imaging techniques such as magnetic resonance (MR) and/or computerised axial tomography (CT) were used for cases with media opacities or extensive retinal detachment that prevented correct tumour size evaluation. These techniques were also used to detect extraocular extension. Liver ultrasound, Chest X-ray and liver function test were performed to evaluate systemic extension.

The criteria for melanoma diagnosis was the observance of a lesion with ophthalmoscopic appearance and compatible ultrasound characteristics with a height between 1 and 3 mm or a base and a base of more than 5 mm. Based on the criteria published in the "Collaborative Ocular Melanoma Study" (COMS) [13, 14], the melanomas were divided into three sizes: small choroidal melanomas, those with a height ≤3 mm and a base between 5 and 10 mm; medium, those with a height between 3.1 and 10 mm and a base less than or equal to 16 mm; and large, those with over 10.1 mm of height or more than 16 mm of base. In addition, TNM classification [16] was used: T1, tumour ≤10 mm of greatest basal diameter and ≤2.5 mm in greatest height (thickness); T2, tumour >10 but no more than 16 mm in greatest basal diameter and between 2.5 and 10 mm in greatest height; T3, tumour >16 mm in greatest diameter and/or >10 mm in maximum height (thickness); T4, tumour >16 mm in greatest diameter and/or >10 mm in maximum height (thickness), with extraocular extension. NX, no regional lymph node metastasis can be assessed; N0, no regional lymph node metastasis; N1, regional lymph node metastasis; MX, distant metastasis cannot be assessed; M0, no distant metastasis; M1, distant metastasis (Table 1).

The following data were registered for all patients: general demographic and past history patient data and ocular examination; tumour size, shape and location; signs of tumour activity, and signs of systemic and extraocular extension. An ophthalmologist from the unit was in charge of filling in the questionnaire data. The completed questionnaires were codified in a database designed in Microsoft® Access®, after informed patient consent was obtained.

Treatment protocol

Brachytherapy was indicated in medium (T2N0M0) or small sized tumours (T1N0M0) (with documented growth). It was also indicated for large tumours (T3N0M0) in which the patient rejected enucleation or in cases of a single eye affected. The prescription dose in the tumour apex was 100 Gy from September 1997 until January 2003. From then on, it was 85 Gy, in accordance with COMS directives that reduced the dose needed for treatment of these tumours, and coinciding with the acquisition of this type of plaque [7]. The ophthalmoscopic, ultrasound and CT/MR measurements were introduced in the dosimetry programme BEBIG Plaque Simulator™ version 2.16. Australian Radiation Oncology Physics and Engineering Services (ROPES)-type plaques were used between September 1997 and January 2003; COMS-type plaques were used from then until the end of patient inclusion in the study.

Plaque placement and removal

Once treatment indication had been decided, the patients were informed of the process and the different treatment alternatives. After obtaining written patient consent and performing dosimetry, each patient was admitted to the hospital; in a programmed operation, the plaque was implanted with a 2-mm tumour base margin. Next, the patient was placed in an isolated room, remaining there for the number of days required to liberate the needed dose according to the dosimetric calculation. After that period, the patient was transferred to the operating room and the plaque was removed under local anaesthesia.

Result evaluation and follow-up

All patients were followed up in the Ocular Oncology Unit one week after the treatment and then at one, three, nine and twelve months p.o. to evaluate any appearance of early complications and to establish treatment efficacy objectively. At these visits, VA was measured and ultrasound and fundus eye exams were performed. The treatment was considered effective if the tumour had stopped growing in successive ultrasonography controls or had decreased in size [9]. After that point, follow-ups were performed every six months over the first two years and each year after that to confirm local tumour control, and to prevent and treat the appearance of late treatment side effects and detect any metastasis. In addition to the previously mentioned ophthalmologic exams, a liver ultrasound and liver function test were performed once a year in the follow-ups. The data obtained in each follow-up were noted on a form and introduced in the database mentioned earlier.

Statistical treatment

A historical cohort study was performed. The data are represented with the mean and standard deviation or in frequency tables, as applicable. VA and maximum ultrasound height throughout the study were compared by the nonparametric Friedman test for related samples. Values of $p<0.05$ were considered statistically significant. The statistical program SPSS version 12.0 was used for the statistical analysis.

Results

From the total of 310 patients diagnosed with uveal melanoma during the study period, 136 underwent EB with ^{125}I seeds, with a mean follow-up of 55.3 months (SD=31.6). For 94.1% of the patients, the follow-ups were for over 1 year, 41.4% four years and for 30.1% more than 5 years. There were 7 lost patients in follow-up (patient did not attend clinic for more than 1 year).

Table 2 Baseline patient and tumour characteristics

	136	%
Mean age (years): 58.3 (SD=13.2)		
Mean follow-up (months): 55.3 (SD=31.6)		
Gender		
Female	65	47
Male	71	53
Laterality		
Right	64	47
Left	72	53
Tumour size		
Small	10	5.7
Medium	110	80.9
Large	16	11.8
TNM		
T1	6	4.4
T2	121	89
T3	8	5.9
T4	1	0.2
Tumour shape		
Nodular	91	66.9
Mushroom	41	39
Flat	4	2.9
Follow-up		
>1 year	128	94.1
>2 years	101	74.3
>3 years	73	53.7
>4 years	60	44.1
>5 years	41	30.1

Greatest thickness: 5.8 (SD=2.4)
Maximum basal diameter: 11.4 (SD=2.6)

Tumour and demographic characteristics

The general patient and tumour characteristics are presented in Table 2. The mean age at diagnosis was 58.3 years (SD=13.2), without differences with respect to laterality or gender. Of the tumours treated, 80.9% were of medium size. According to the TNM Classification, 89% were T2N0M0, the majority being nodular (66.9%) or mushroom shaped (39%). No tumours with extraocular extension (T4) were treated, except for one case in which the extension was small and found at the base of the tumour within the field of action of the brachytherapy (Table 2).

Treatment characteristics

Patients treated until January 2003 with ROPES plaques at a dose of 100 Gy received a mean prescribed dose liberated in the tumour apex of 92.6 Gy, and those treated after that date with COMS plaques at a dose of 85 Gy received a mean dose of apex 89.6 Gy (SD=7.25). Treatment time ranged from 2 to 8 days, depending on the seed charge and tumour size, the mean treatment length being 5 days.

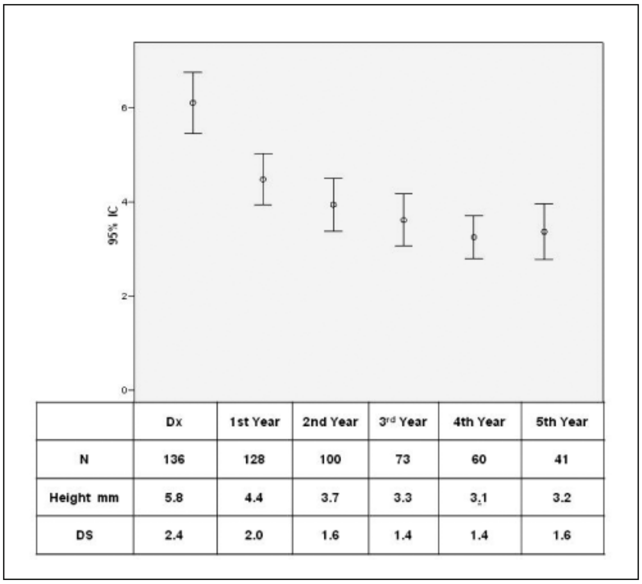

Fig. 1 Evolution of tumour height

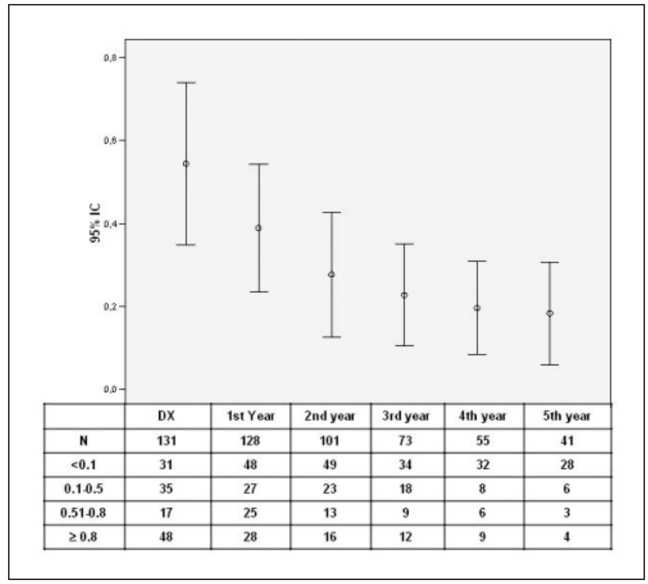

Fig. 2 Evolution of VA

Tumour control

In Fig. 1, the evolution of tumour height is presented. This was measured by ultrasound and validated by the nonparametric Friedman test for related samples, showing a statistically significant decrease (p<0.005). It can be seen that, at the fifth year of follow-up, the tumours presented an average height reduction of 57.9%. A tumour relapse after treatment was observed in four cases.

Visual acuity evolution

Figure 2 shows the evolution of VA validated by the nonparametric Friedman test for related samples (p<0.005). It can be seen that, at the moment of diagnosis, almost 50% of the patients had a VA equal to or greater than 0.5. In the patients who had five years of follow-up, 16.2% maintained VA and 17.2% improved.

Appearance of side effects

In Table 3, the side effects that appeared over the study period are presented. The most prevalent early side effects were cataracts and retinal detachment, while the most prevalent late side effects were radiation-induced retinopathy and optic neuropathy. Retinopathy appeared in 11.3% of the patients in the first year, reaching 17.4% in the third year of follow-up; this percentage remained more or less stable until the end of the follow-up period. Radiation-induced optic neuropathy appeared in approximately 3% of the patients in the first year. It increased to 11.5% in the third year and reached 4.8% in the fifth year.

Eye conservation

Enucleation was performed in ten cases (7.3%), four due to tumour relapse and six because of complications

Table 3 Side effects

	1st year		2nd year		3rd year		4th year		5th year	
	N=132	%	N=116	%	N=95	%	N=77	%	N=62	%
Radiation retinopathy	15	11.3	26	22.4	23	24.2	12	15.5	12	19.3
Optic neuropathy	4	3	9	7.7	11	11.5	9	11.6	3	4.2
Cataract	19	14.3	19	16.3	23	24.2	20	25.9	17	27.4
Diplopia	2	1.5	0	0	0	0	2	2.5	0	0
Retinal detachment	42	31.8	16	13.3	7	7.3	7	9	5	8
Pain	6	4.5	0	0	4	4.2	4	5.1	1	1.6
Vitreous haemorrhage	10	7.5	5	3.7	3	3.1	3	3.8	4	6.4
Scleral necrosis	2	1.5	1	0.8	0	0	1	1.2	2	3.2
Neovascular glaucoma	4	3	1	0.8	3	3.1	1	1.2	1	1.6
Ptosis	1	0.7	0	0	2	2.1	0	0	0	0

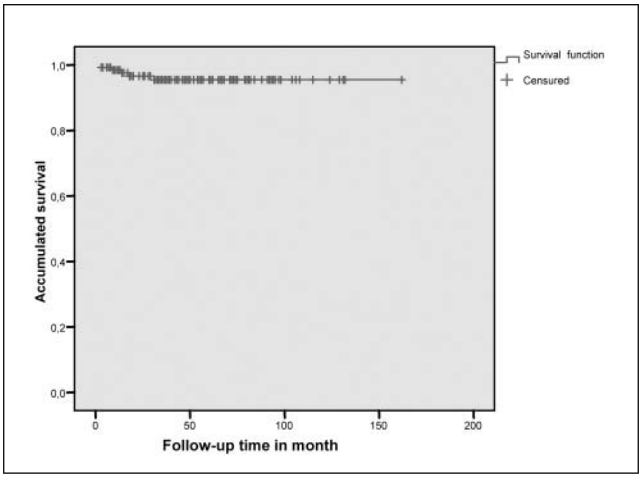

Fig. 3 Survival function

from the radiation. The percent of eye conservation was 92.6%.

Survival

Mean survival was 155.4 months (SD=2.8). Three patients died during the course of the study, but only one was due to the melanoma. The overall survival at 13 years is therefore 97.1% (Fig. 3).

Discussion

The objectives in conservative treatment of choroidal melanoma are, in the first place, the destruction of the tumour, preservation of the eye and, insofar as possible, visual function. The results from EB in this study, with a mean follow-up of approximately 5 years, showed a disease control above 97%, given that only 4 cases of relapse were observed; a mean height reduction at 5 years of 55.1% was achieved. Caminal et al. [17] published similar data, with a reduction of 53.84% in 55 patients with a follow-up of more than a year. The Collaborative Ocular Melanoma Study Report No19 (COMS) provided data on 650 patients; of those patients, 411 had been followed up for over 5 years; 57 of these presented relapses and were enucleated (8.7%) [18]. These data are similar to those published for other isotopes used previously, such as iridium 192 [19] and ruthenium 106 [20], which achieved regression or stabilisation in over 90% of the cases.

In this study, eye conservation was over 93%. During the period the study lasted, 10 eyes were enucleated, 4 for tumour relapse and 6 due to complications. In Caminal et al.'s [17] series of 55 patients, 6.8% were enucleated; half of them because of recurrence and half due to complications. Correa et al., in a recent paper [21] studying 120 patients, report a local control at 5 and 8 years of 88.2% and 72.7%. This data may suggest the longer follow-up may achieve a higher number of relapses. COMS provided figures for patients with 5-year follow-ups of 0.6% secondary enucleation, most due to tumour recurrence [22]. Our indexes of relapse and enucleation secondary to complications seem to be smaller than those previously published.

With respect to conservation of VA, the levels were low at the beginning of this study. This was because approximately 50% of the patients presented V <0.5 at the moment of diagnosis due to the associated tumour pathology (retina detachment or oedema, macular degeneration from the tumour). At any rate, as can be seen in the data presented, VA decreased inexorably as follow-up time advanced. Somewhat more than 40% of the patients treated had VA above 0.1 in the third year of follow-up, while only 30% reached this level in the fifth year. Similar data are presented by other authors. Caminal et al. [17], with a 1-year follow-up, found 32.7% of the patients with VA >0.1 at their last follow-up. Shields et al. [15] analysed the VA development of 1106 patients treated with EB, finding that VA was less than 0.1 in 34% of their patients at 5 years, and in 68% at 10 years of follow-up. COMS [22] analysed this variable in their Report No. 16, finding in a 3-year follow-up that 49% of their patients experienced a significant reduction in their vision, 43% having VA <0.1.

The side effects that appear most frequently are cataract, radiation-induced retinopathy, neuropathy and retinal detachment. Cataract is an early treatment complication that usually appears in the first year of follow-up, due to the great sensitivity of the lens to radiation. Retinopathy and optic neuropathy secondary to radiation are complications with a marked tendency to appear late. Their incidence therefore increases as follow-up time lengthens. The incidences of retinal detachment shown in Table 3 were present in the majority of the cases at the moment of diagnosis. This decreased considerably in the first and second years of follow-up as a consequence of tumour control.

With respect to survival, only one patient died from liver metastasis. However, one of the limitations of this study is the length of follow-up, given that patients with metastasis appear in the series published over the course of time and that a more prolonged patient follow-up is needed for their evaluation.

The results of this study confirm that iodine 125 EB is an effective treatment in the control of uveal melanoma, which manages to preserve VA and the eye in the majority of the cases.

Conflict of interest The authors declare that they have no conflict of interest relating to the publication of this manuscript.

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